

On the Study of Different & Rotor Geometry Configuration for use in High-Speed Induction Motors

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Abstract

Abstracts contain most of the following kinds of information in brief form. The body of your paper will, of course, develop and explain these ideas much more fully. As you will see in the samples below, the proportion of your abstract that you devote to each kind of information and the sequence of that information will vary, depending on the nature and genre of the paper that you are summarizing in your abstract. And in some cases, some of this information is implied, rather than stated explicitly. The Publication Manual of the American Psychological Association, which is widely used in the social sciences, gives specific guidelines for what to include in the abstract for different kinds of papers for empirical studies, literature reviews or meta-analyses, theoretical papers, methodological papers, and case studies.

Contents

1 Solid Rotor Technology	1
1.1 Introduction	4
1.2 Theory and Modeling of a Solid-rotor IM	5
1.3 Calculations of Main Dimensions	5
1.3.1 Calculations of Rotor Diameter	6
1.3.2 Calculations of Active Rotor Length	7
1.3.3 Solid-rotor Motor with Sine-wave Supply	7
1.4 Determination of Equivalent Circuit Parameters	8
1.5 Calculation of Skin Depth	9
1.6 Effect of End- Rings	10
1.6.1 Calculation of Russel Factor	10
1.6.2 Calculation of Trickey Factor	11
1.7 Types of Solid-rotor Types	11
1.7.1 Smooth-Solid Rotor induction Motor	11
1.7.2 Axially-Slitted Solid Rotor Induction Motor	12
1.7.3 Copper Coated Solid Rotor Induction Motor	12

List of Figures

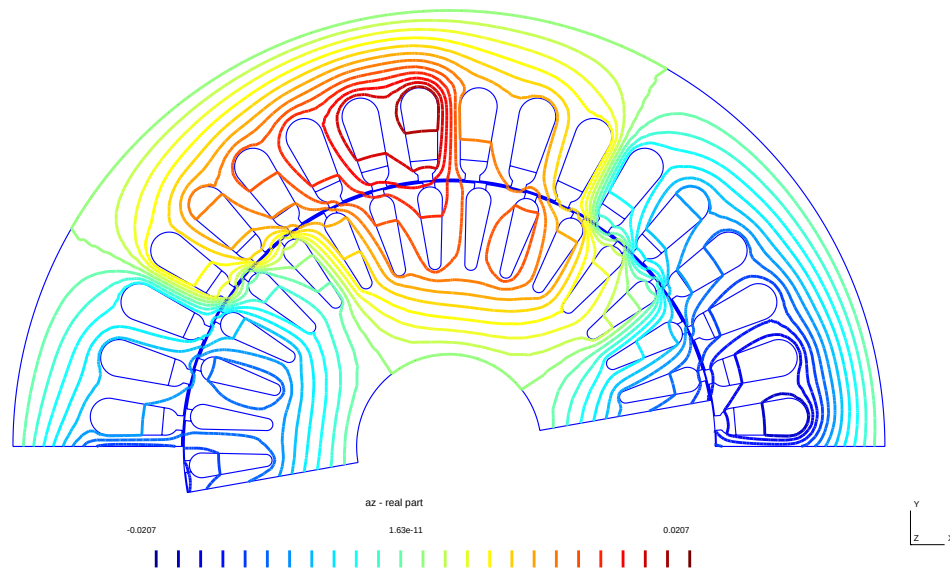
List of Tables

Solid Rotor Technology

In industrial settings, the Polyphase induction motors (IMs) have shown clear advantages due their efficiency and construction simplicity as over more than a century of work and research went into improving its overall performance. These structural optimization along with the industries lack of need for constant torque output in the industry made IMs the preferred choice. Furthermore their most distinctive characteristic is, by using different types of rotor construction, different types of torque characteristics can be achieved.

The most common type of IM is the Squirrel-Cage IM (CRIM). Such a motor consists of a steel cylinder, with the slots being filled with mostly aluminum conductors. The rotor structure is laminated to decrease the eddy currents that are being induced in the rotor. These currents are the main cause of eddy current losses in a rotor. The skewed body structure also helps the rotor to avoid the cogging torque at its start of operation. The skewed design also indirectly help in the increase of the total rotor impedance, which in turn increase the starting torque.

Although a cage-induction machine has widespread use in industrial applications, the weakness of its structure for high-speed implementation limits its use to the current range of applications. Depending on the structural complexity, the rotor design has several constraints that limit its operation in terms of angular speed, since the rotor cannot withstand the large rotational stresses caused by the high circumferential velocity of the rotor. Practical studies have shown that a squirrel-cage rotor



cannot operate at speeds of 50,000 rpm [1].

Even though squirrel-cage IM is not possible to use in high-speed application various alternatives have been proposed. One of the most promising one is to use a solid body equipped with a squirrel cage [22,23,24]. A solid body with a squirrel cage is made from a ferromagnetic material such as Fe52 (S355J0/EN 10025), while the squirrel cage is made from copper alloy. However the construction of this type of rotor is complicated and with the presence of various parts it is not a popular choice in high-speed applications.

1.1 Introduction

When a high-speed application is targeted, a solid rotor is by far the better choice. The motors with solid rotors have advantages over other types of induction motor in terms of mechanical strength and structural simplicity, as well as being easier to manufacture [25]. Solid-rotor Induction Motor is built from a rotor made of a single piece of ferromagnetic material. For this reason, solid rotors

are superior to conventional cage-rotor structures in terms of their ability to operate in extremely high-speed applications and their durability with respect to high thermal stresses that ensure low levels of vibrations under all operating conditions [20].

As it is known the construction of a CRIM requires complex production methods. By developing solid rotors we can overcome the complexity of IM manufacturing. Apart from the advantages of the SRIM mentioned above, the others are given below. These are mainly;

- If the rotor is a SSRIM motor, because of the absence of slots there will be no vibrations or noise
- High mechanical integrity, reliability and durability
- The possibility of obtaining steady-state stability
- Linearity of torque-speed characteristics.

While the reasons aforementioned above show us the great benefits of this technology as is with other motor types there are some disadvantages. These are mainly; lower output power (P_{out}), efficiency (η) and power factor ($\cos\phi$), higher load slip compared to CRIM and high rotor impedance (Z_{Rotor}). However these disadvantages can be decreased or diminished by the following ways;

1. The effects of high rotor impedance can be controlled or its effect maybe diminished by using an optimum control system.
2. A coated (layered) structure can be used with appropriate materials made from ferromagnetic and nonmagnetic high-conductivity materials.
3. A ferro-magnetic material for the rotor with its magnetic permeability to its electric conductivity as small as possible

1.2 Theory and Modeling of a Solid-rotor IM

A solid-rotor induction motor operates according to the same principles of operation as a conventional induction motor. The performance of such a motor is characterized by the nature of the interaction between the air-gap revolving field and the eddy currents induced in the solid-rotor, which develop the electromagnetic torque. It is necessary to evaluate the impedance of the solid-rotor, both in phase and magnitude, to predict the behaviour of the motor under load conditions. The main difficulty in deriving an expression for this impedance arises from the extremely nonlinear magnetisation characteristic of the steel material. Many different approaches have been adopted to the calculation of eddy current loss in unlaminated magnetic materials aiming to develop an equivalent rotor impedance to be included in the parameters of motor equivalent circuit.

Early attempts are based on the assumption of material constant permeability, i.e. they considered the rotor as a linear medium. More realistic attempts used non-linear approximations to fit the magnetising curve, such as a limiting non-linear rectangular approximation. The limiting non-linear method is well established and its result agrees well with practical measurements. It is simple to use, and widely-accepted since the magnetising force at the surface of the solid-rotor is usually high enough to drive the rotor material well into saturation. Both the non-linear and linear models have found application in the modeling of a solid-rotor machines.

1.3 Calculations of Main Dimensions

Even though when designing the high-speed rotor that is manufactured from a solid piece of steel, the effects of centrifugal force must be taken into account. The force that acts on the rotor while its active must be kept at minimum. To compare this force, a high speed rotor experiences the same centrifugal force as a large generator. As an example would be a 300 mm high speed rotor with 10,000 rpm has a surface speed of 157 m/s. This is the same speed as a 2000 mm generator with 1500 rpm generator experiences. This in the end forces the design of the rotor by the mechanical forces it experiences.

The mechanical strength of electrical steel limits the laminated rotors surface speed to about 150 m/s. If Cobalt-iron laminations were used, a higher speed could be obtained. But the price of this material makes it a seldom choice.

1.3.1 Calculations of Rotor Diameter

When designing a solid rotor, the mechanical tangential stress affecting the rotor is a primary parameter in the determination of the output torque of the motor. The torque T can be calculated from the electro-magnetic rotor surface tangential stress σ_{tan} ,

$$T = \sigma_{tan} 2\pi r^2 L_r \quad (1.1)$$

Where L_r and r are the length and the radius of the rotor, respectively. Equation (2.1) shows that the diameter value of the rotor has a strong effect on the output torque. The increase of the rotor diameter can give a higher torque output if the tangential stress kept constant. This creates a limit on rotor diameter. A solid rotor construction has to withstand high mechanical stresses. According to Ylinen, the mechanical stress on a cylindrical solid rotor can be estimated by,

$$\sigma_{max} = C \rho r^2 \omega^2 \quad (1.2)$$

Where ρ is the mass density of the material and ω is the angular velocity of the rotor. According to analysis, the maximum stress of a smooth homogenous cylinder takes place at the cylinder centre. If the center is hollow, the maximum stress will be experienced at the inner radius of the cylinder. The C factor can be calculated by; 11

$$C = \frac{3 + \nu}{8} \quad (1.3)$$

$$C = \frac{3 + \nu}{4} \quad (1.4)$$

$$C \approx 1 \quad (1.5)$$

Where (2.3) is for smooth homogenous cylinder, (2.4) is for a cylinder with a small bore and (2.5) is for a thin cylinder. ν is the Poisson's coefficient for the material that is being used.

1.3.2 Calculations of Active Rotor Length

The length of the rotor is heavily influenced by the natural frequencies of the rotor. If the rotor is operated at critical speeds, the high vibrations it experiences will damage the rotor. Because of its solid body, Solid rotors possess the highest critical frequencies in a electric rotor.

The rigidity of the rotor is heavily influenced on the elasticity, density and the geometry of the rotor. Young's modulus E describes the elasticity of the core material. According to Wiat, the length of the rotor is,

$$L = \sqrt{n^2 \frac{\pi^2}{k\Omega} \sqrt{\frac{El}{\rho A}}} \quad (1.6)$$

Where A is the cross-section of the cylinder and n is the ordinal of the critical speed and 1 is the modulus of the inertia. k is the ratio between the n th critical frequency and the nominal angular velocity

$$k = \frac{\omega_c}{\omega_n} \quad (1.7)$$

1.3.3 Solid-rotor Motor with Sine-wave Supply

In a Solid-rotor, the flux penetration mechanism into the magnetic material is heavily depends on the magnetic nonlinearity of the iron. With that in mind, To model an equivalent circuit for a solid-rotor, the parameters numerical values depends strongly on the magnitude of the voltage applied to the stator terminals. 12 When designing a equivalent circuit, the input voltage was considered as pure sinusoidal. The flux penetration into a solid steel assumes that the flux density in the steel rotor structure may only exist at a magnitude equal to the saturation level $\pm B_s$. Thus for a given ϕ , as approximately occurs only when constant voltage is applied. δ is the air gap is the air gap and constant for any given rotor frequency [26]. By using the limiting non-linear representation in the analysis of solid-rotor gives an expression for the equivalent rotor impedance referred to the stator. From this, the general expression of rotor impedance is given as;

$$Z_{rotor} = \left(\frac{AmL^2N^2\rho B_s}{K_s D \phi_s} \right) \angle \theta \quad (1.8)$$

Where A is the calculation constant, m is the number of stator poles, N is the effective number of stator turn per phase, ρ is the rotor resistivity, B_s is the saturation flux density of the rotor material, K_e is the end-effect factor, D is the rotor diameter s is the slip and θ is the constant for empirical adjustment.

1.4 Determination of Equivalent Circuit Parameters

The equivalent circuit of a polyphase-IM with a solid-rotor body is given in Figure 4, where r_1 and x_1 represents the stator winding resistance and the leakage reactance X_m represents the magnetizing reactance and Z is the rotor impedance referred to the stator at the fundamental supply frequency. In this model the core losses are neglected. The equivalent impedance per phase is given by;

$$Z = Z_1 + \frac{Z_m Z_{rotor}}{(Z_m + Z_{rotor})} \quad (1.9)$$

Where;

$$Z_1 = r_1 + jx_1 \quad (1.10)$$

$$Z_m = jX_m \quad (1.11)$$

The input power to the rotor per phase is given by; 13

$$P_2 = I_2^2 R_{rotor} \quad (1.12)$$

In which R_{rotor} represents the real part of the rotor impedance Z_{rotor} . The developed gross output power per phase is;

$$P = (1 - s)P_2 \quad (1.13)$$

The developed total torque is;

$$T = 3 \frac{P}{\omega_r} \quad (1.14)$$

$$\omega_r = (1 - s)\omega_s \quad (1.15)$$

Where ω_s represents the synchronous speed of the stator field. I_1 I_2 X_{12} r_1 I_m jx_m
Z V Figure 2. 2. Single-phase equivalent circuit of a Solid-rotor IM

1.5 Calculation of Skin Depth

When designing a solid-rotor for high-speed applications, special care must be made to optimize its electromagnetic properties. Solid-rotor structures are heavily influenced by the eddy currents since the rotor is a single piece of ferromagnetic material. According to Lenz's law, eddy currents create a magnetic field that goes against the initial field that created it.

This in turn causes the so called "skin-effect" on the surface, where the magnetic flux lines are pushed to the surface of the rotor. According to [26] the depth of the first harmonic magnetic field is;

$$\delta_i = \left[\operatorname{Re} \left(\sqrt{\frac{1}{\left(\frac{\pi}{\sigma}\right) + j \cdot 2 \cdot \pi \cdot f_s \cdot s \cdot \frac{\sigma_{steel}}{K} \cdot \mu_0 \cdot \mu_r}} \right) \right]$$

Where f_s is the frequency of the rotating magnetic field, s is the slip of the rotor, τ is the pole pitch, σ_{steel} is the electric conductivity of steel, μ_0 is the permeability of vacuum, μ_r is the permeability of steel and K_t is the correction factor. The equation can further be simplified by substituting;

$$x = \left(\frac{\pi}{\tau}\right)^2 \quad (1.16)$$

$$y = j \cdot 2 \cdot \pi \cdot f_s \cdot s \cdot \frac{\sigma_{steel}}{K_r} \cdot \mu_0 \cdot \mu_r \quad (1.17)$$

From the equations mentioned above we can obtain a simpler definition for the skin effect depth;

$$\delta_i = \frac{\sqrt{x + \sqrt{x^2 + y^2}}}{\sqrt{2(x^2 + y^2)}} \quad (1.18)$$

The reciprocal of correction factor is;

$$K_T = \frac{1}{\tanh\left(\pi \cdot \frac{L}{2\tau}\right)} \geq 1$$

$$1 - \frac{\tanh\left(\pi \cdot \frac{L}{2\tau}\right)}{\left(\pi \cdot \frac{L}{2\tau}\right)} \quad (1.19)$$

Where L is the length of the rotor.

1.6 Effect of End- Rings

When doing 2D analysis of a 3D problem, the system must be remodelled to calculate the properties. By recalculating the resistivity of the rotor, the rotor end effects can be calculated. 15
$$I_{Fe} < /math > I_{Fe}$$
 Figure 2. 3. The Effects of End-Rings on rotor current
To do this, the rotor equivalent resistivity ρ must be modified by an end-effect factor k ;
$$\rho_{new} = \frac{\rho}{k} < /math > (2.21)$$

1.6.1 Calculation of Russel Factor

According to [27], the rotor end can be calculated in the simulation by modifying the rotor resistivity by the end-factor depending on the geometry of a solid rotor;

$$k_{Russel} = 1 - \frac{2\tau_p}{\pi L_r} \tanh \left(\frac{\pi L_r}{2\tau_p} \right) \quad (1.20)$$

Where τ_p is the pole pitch and L_r is the rotor length.

1.6.2 Calculation of Trickey Factor

According to [28], the end-factor depends on the inner and the outer diameter and the pole-pair number of the rotor, which is defined as;

$$k_{Trickey} = \frac{p}{2} \left| \frac{1 + \left(\frac{D_{in}}{D_r} \right)^p}{1 - \left(\frac{D_{in}}{D_r} \right)^p} \right| \left(1 - \frac{D_{in}}{D_r} \right) \quad (1.21)$$

Where D_{in} is the inner rotor diameter and p is the pole pair number. In the case of a slitted rotor, the inner diameter D_{in} can be defined as; 16

$$D_{in} = D_r - 2b_{slit} \quad (1.22)$$

Where b_{slit} is the depth of the rotor slit. It should be said that Trikey's formula does not take the rotor length into account. Thus, according to Trickey, the correction factor is independent of rotor length and constant for all rotor length. 2.8.3 Calculation of Yee Factor According to [29], it is assumed that the rotor current density is confined in a thin shell that covers the rotor. The end-factor is calculated as;

$$k_{Yee} = \frac{aL_r \left(1 + \coth\left(\frac{aL_r}{2}\right)\right)}{aL_r \left(1 + \coth\left(\frac{aL_r}{2}\right)\right) - 2} \quad (1.23)$$

Where $a = \pi/\tau_p$ and L_r is the rotor length.

1.7 Types of Solid-rotor Types

1.7.1 Smooth-Solid Rotor induction Motor

The simplest rotor type for solid rotors is the smooth, steel cylinder. [1] It is cheap and easy to manufacture such a solid rotor because of its basic geometrical shape. The simple structure means it has the best mechanical, fluid and thermal dynamics possible in an electrical rotor when compared to other rotors [18]. However, its electromagnetic properties are not as good as the other types, for example, its slip value is much higher for same load torque. As a result of the skin effect, eddy currents travel around the surface of the rotor, and travelling eddy currents at the surface of a ferromagnetic rotor material push the induced magnetic field towards the outer layer of the rotor. This means that the depth of the magnetic flux's penetration into the rotor is very shallow. The magnetic flux and the torque-producing eddy currents are concentrated on the surface layer and the interior part of the rotor becomes electrically inert [20].

1.7.2 Axially-Slitted Solid Rotor Induction Motor

When designing a solid rotor, special care must be paid to eliminating the air-gap harmonics that contribute a major part to the rotor losses, predominantly on the

surface of the rotor [18]. In order to minimize the rotor losses, there are different rotor designs. One of the most common rotor types is the axially slitted rotor, which helps to decrease the effects of the rotor losses caused by the air-gap harmonics [20]. The fundamental component of the magnetic flux penetrates deeper into the rotor, thereby improving the electromagnetic performance of the motor in this slitted geometry. In order to obtain the optimum electromagnetic performance with the slitted rotor, the slit width must be 1.5 mm and the slit depth must be half of the rotor's radius [19]. The main drawbacks of an axially slitted rotor are a reduction in the mechanical strength and an increase in the audible noise, the manufacturing costs and the air-friction losses. Since these frictional losses increase as a square of the speed, a smooth cylinder is the most favorable choice for high-speed drive applications [20].

1.7.3 Copper Coated Solid Rotor Induction Motor

The other method to reduce rotor losses is using a conductive coating around the rotor, acting as a low-pass filter for the air-gap harmonics in a high-speed drive application. The coating decreases the impact caused by the air-gap harmonics, while increasing the motor's electromagnetic performance. It has been shown that to obtain the optimum electromagnetic performance with a coated rotor, the thickness of the coating material must be around 1.5 mm, irrespective of the rotor's diameter [21]. This thickness is enough to provide the magnetic benefits, while a thicker coat would adversely affect the motor's performance. In addition, the conductivity of the coating